

A
FIELD PROJECT REPORT
ON
ANALYZING INTERNAL HOSPITAL OPERATIONS
USING BUSINESS ANALYTICS:
A CASE STUDY OF
IMPERIAL HOSPITAL

SUBMITTED BY
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UNDER THE GUIDANCE OF
PROF. VANDANA TRIPATHI

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DECLARATION

I, Utkarsha Wanjare, hereby declare that the Field Project Report entitled “Analyzing Internal Hospital Operations Using Business Analytics: A Case Study of Imperial Hospital” submitted to Savitribai Phule Pune University, Pune, is a record of original work carried out by me under the guidance of Prof. Vandana Tripathi.

This report is submitted in partial fulfilment of the requirements for the award of the degree of Master of Business Administration (MBA).

I further declare that this work has not been submitted to any other University or Institution for the award of any degree, diploma, or similar qualification. The conclusions drawn in this report are based on the data collected and analysed by me.

Place: Balewadi, Pune

Date: _____

Signature of the Student

Utkarsha Wanjare

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I take this opportunity to express my sincere gratitude to Dnyansagar Institute of Management and Research for providing me with the opportunity to undertake this Field Project and for creating an environment that encourages learning and growth.

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I also thank my friends and well-wishers for their encouragement and moral support. Lastly, I express my gratitude to Almighty God for His blessings and strength throughout this journey.

Place: Balewadi, Pune

Date: _____

Utkarsha Wanjare

COMPANY CERTIFICATE



Date- 06-Feb -2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Miss.Utkarsha Prabhakar Wanjare, a student of Dnyansagar Institute of Management and Research has successfully completed field work at Imperial Multispecialty Hospitals for duration of 160 hours.

During the tenure, the student worked on the topic **Analysing Internal Hospital Management Using Business Analytics** under guidance of our team.The student displayed dedication, professionalism and keen interest in learning.

We appreciate her contributions and wish her all the very best in her future academic and professional endeavors.


For Imperial Multispecialty Hospitals
Dr. Pratikha Chaudhary
Medical Superintendent

INSTITUTE CERTIFICATE

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ABSTRACT

The healthcare industry plays a crucial role in delivering efficient and timely services to patients. Effective internal management of hospital operations is essential to ensure quality care, optimal resource utilization, and patient satisfaction. This project, titled “Analyzing Internal Hospital Operations Using Business Analytics: A Case Study of Imperial Hospital, Chikhali”, focuses on examining key operational areas within a hospital using business analytics techniques.

The study aims to analyze various aspects such as patient flow (OPD and IPD), waiting time, admission and discharge processes, staff workload management, billing operations, and resource utilization including beds, medicines, and equipment. Data for the study was collected through structured questionnaires from doctors, nurses, and administrative staff to understand real-time operational challenges.

The collected data was organized and analyzed using tools such as Microsoft Excel and Power BI to identify patterns, inefficiencies, and key problem areas. The analysis highlights issues such as patient delays due to high patient load, staff shortages, coordination gaps, and limitations in existing systems.

Based on the findings, the study suggests improvements in staff management, process optimization, digital system enhancement, and better coordination between departments. These recommendations aim to improve operational efficiency, reduce patient waiting time, and enhance overall hospital performance.

This project demonstrates how business analytics can be effectively used in healthcare management to support decision-making and improve service quality.

CHAPTER NO. 01

INTRODUCTION TO PROJECT

Introduction

In today's healthcare environment, hospitals handle a large number of patients, medical records, staff activities, billing processes, and resource management tasks on a daily basis. Managing these operations efficiently is essential for providing quality healthcare services and improving patient satisfaction. However, many hospitals face operational challenges such as long patient waiting times, staff workload imbalance, delays in admission and discharge procedures, shortage of beds and equipment, and lack of coordination among departments. These issues can directly affect hospital performance and patient experience.

Business Analytics has emerged as an effective approach for analyzing hospital operations and improving decision-making processes. By using analytical tools and data-driven techniques, hospitals can identify operational inefficiencies, monitor performance, optimize resource utilization, and improve overall workflow management. Business Analytics helps healthcare organizations convert raw operational data into meaningful insights that support strategic planning and better service delivery.

This project, titled **“ANALYZING INTERNAL HOSPITAL OPERATIONS USING BUSINESS ANALYTICS: A CASE STUDY OF IMPERIAL HOSPITAL, CHIKHALI”**, focuses on studying and analyzing the internal operational activities of Imperial Hospital. The study examines various operational areas such as patient flow management, OPD waiting time, admission and discharge processes, staff workload management, technology usage, inventory and bed management, and overall hospital efficiency.

The project is based on both primary and secondary data collection methods. Primary data was collected through surveys and questionnaires from doctors, nurses, and administrative staff to understand real-time operational challenges within the hospital. The collected data was analyzed using tools such as SQL, Excel, Python, and Power BI to generate dashboards, reports, charts, and meaningful business insights.

The main objective of this study is to identify operational gaps, improve hospital workflow efficiency, and provide data-driven recommendations for better hospital management. The study demonstrates how Business Analytics can support healthcare institutions in improving operational performance, resource planning, patient care services, and overall organizational effectiveness.

1.1 Statement of the Problem

Hospitals today operate in a highly demanding and complex healthcare environment where efficient management of internal operations is essential for delivering quality patient care. With the increasing number of patients, growing healthcare expectations, limited medical resources, and continuous operational pressure, hospitals often face significant challenges in maintaining smooth and effective workflows. Inefficiencies in operational processes can directly impact patient satisfaction, staff productivity, resource utilization, and the overall performance of the hospital.

Several operational activities such as patient registration, appointment scheduling, consultation, admission, billing, discharge procedures, inventory handling, and bed allocation require proper coordination among different hospital departments. However, delays and inefficiencies frequently occur due to high patient load, manual processes, communication gaps, insufficient staffing, and lack of process standardization. These issues often result in increased waiting times, overcrowding, delayed treatment, staff stress, and reduced operational efficiency.

One of the major concerns in hospital management is the effective utilization of human resources. Problems such as staff shortages, uneven workload distribution, poor shift planning, and lack of training can affect the productivity and performance of doctors, nurses, and administrative staff. During peak operational hours, hospitals may struggle to manage patient flow efficiently, leading to work pressure and reduced service quality.

In addition to workforce-related challenges, hospitals also face difficulties in managing critical resources such as ICU beds, medicines, medical equipment, and patient records. Delays in discharge procedures, bed cleaning, billing clearance, and interdepartmental coordination can create bottlenecks in bed availability and patient management. Improper inventory management and equipment shortages may further impact hospital operations and patient care services.

Technology also plays a vital role in modern healthcare management. However, many hospitals still rely partially on manual systems or outdated digital systems that limit operational efficiency and data accessibility. Software-related issues, incomplete digitization, and inefficient data management can reduce the effectiveness of hospital operations and decision-making processes.



Although hospitals generate large volumes of operational data every day, this data is often not analyzed effectively to identify trends, monitor departmental performance, improve workflow, or support strategic decisions. Without proper analysis, hospitals may fail to recognize operational gaps and opportunities for improvement. This creates a significant gap between existing hospital

operations and optimal performance standards.

Therefore, there is a need to apply Business Analytics techniques to study and improve internal hospital operations. This study, titled “ANALYZING INTERNAL HOSPITAL OPERATIONS USING BUSINESS ANALYTICS: A CASE STUDY OF IMPERIAL HOSPITAL, CHIKHALI”, aims to analyze various operational aspects of Imperial Hospital including patient flow, staff workload, resource management, operational efficiency, and technology utilization. The study seeks to identify operational inefficiencies, understand their root causes, and provide data-driven recommendations to improve hospital performance, resource optimization, workflow management, and quality of healthcare services.

1.2 Purpose / Objectives of the Project

Purpose

The purpose of this study is to analyze the internal hospital operations of Imperial Hospital using Business Analytics techniques in order to identify operational inefficiencies and improve overall hospital performance. The study focuses on understanding how various operational activities such as patient flow, admission and discharge procedures, staff workload management, resource allocation, billing processes, and technology usage affect the efficiency of hospital operations.



The study also aims to examine the challenges faced by doctors, nurses, and administrative staff in managing daily hospital activities. By collecting and analyzing operational data, the research seeks to identify delays, coordination gaps, workload imbalance, and resource shortages that impact patient care and service quality.

Through the use of analytical tools such as SQL, Excel, Python, and Power BI, the study demonstrates the importance of Business Analytics in healthcare decision-making and operational improvement. The findings of the study will help provide practical recommendations for improving hospital workflow, resource utilization, staff productivity, and patient satisfaction.

Purpose of the Study – Key Points

- To analyze the internal operational processes of the hospital.
- To study patient flow management and waiting time issues.
- To identify challenges in admission and discharge procedures.
- To evaluate staff workload and shift management efficiency.
- To examine the utilization of hospital resources such as beds, ICU facilities, medicines, and equipment.
- To understand the impact of technology and digital systems on hospital operations.
- To identify operational inefficiencies and their root causes.
- To apply Business Analytics techniques for data analysis and decision-making.
- To generate meaningful insights using SQL, Excel, Python, and Power BI.
- To provide data-driven recommendations for improving operational efficiency and service quality.
- To improve coordination, workflow management, and resource optimization within the hospital.
- To enhance overall patient care and hospital performance.

Objectives of the Study

The main objectives of this project are as follows:

- To analyze the internal hospital operations of Imperial Hospital using Business Analytics techniques.
- To study patient flow management and identify delays in registration, consultation, admission, and discharge processes.
- To evaluate staff workload, shift management, and operational efficiency within the hospital.
- To identify operational challenges faced by doctors, nurses, and administrative staff.
- To examine the utilization and management of hospital resources such as beds, ICU facilities, medicines, and medical equipment.
- To analyze the effectiveness of hospital technology systems and digital processes.
- To identify bottlenecks, coordination gaps, and inefficiencies in hospital workflows.
- To use analytical tools such as SQL, Excel, Python, and Power BI for data collection, analysis, and visualization.
- To generate meaningful insights, dashboards, and reports for better decision-making.
- To provide data-driven recommendations for improving hospital operations and service quality.
- To enhance resource utilization, workflow management, and patient satisfaction.
- To improve the overall operational performance and healthcare service delivery of the hospital.

1.3 Theoretical Framework

The theoretical framework of this study is based on the application of Business Analytics concepts and Operations Management principles in the healthcare sector. Hospitals function as

complex service organizations where operational efficiency depends on the proper coordination of people, processes, technology, and resources. Effective management of hospital operations is essential for improving patient care, reducing delays, optimizing resource utilization, and enhancing overall service quality.

One of the major concepts used in this study is **Process Flow Analysis**, which focuses on understanding the movement of patients through different operational stages such as registration, consultation, admission, treatment, billing, and discharge. Inefficiencies at any stage can create bottlenecks, increase patient waiting time, and affect service delivery. By analyzing patient flow, the study aims to identify delays and improve workflow efficiency within the hospital.

Another important concept included in the framework is **Resource Optimization**. Hospitals rely heavily on the effective utilization of resources such as hospital beds, ICU facilities, medicines, medical equipment, and support services. Improper allocation or shortage of resources can negatively affect patient treatment and operational performance. Resource optimization helps in ensuring the maximum utilization of available resources while minimizing operational delays and inefficiencies.

The framework also incorporates **Workforce Management**, which focuses on staff availability, workload distribution, scheduling, and shift management. Doctors, nurses, and administrative staff are essential for smooth hospital operations. Issues such as staff shortages, unequal workload, lack of training, and improper shift planning can reduce employee productivity and service quality. Effective workforce management helps hospitals maintain operational balance and improve staff performance.

Another significant concept used in this study is **Technology Adoption in Healthcare**. Modern hospitals increasingly depend on digital systems and hospital management software for maintaining patient records, billing, scheduling, reporting, and communication between departments. The use of technology improves operational accuracy, reduces manual errors, enhances coordination, and supports faster decision-making. The study examines how digital and manual systems impact operational efficiency within the hospital.

The study is further supported by **Queue Theory**, which is widely used in healthcare operations management to analyze waiting lines and service processes. Queue Theory helps in understanding patient waiting time, service delays, and congestion, especially in OPD departments where patient flow is high. By applying this concept, the study aims to identify factors contributing to long waiting times and suggest improvements in service processes.

In addition, the framework includes **Performance Measurement and Efficiency Analysis**, which focuses on evaluating hospital performance using operational indicators such as patient waiting time, bed occupancy, staff efficiency, workload management, discharge delays, and resource utilization. These performance measures help in identifying operational gaps and monitoring overall hospital efficiency.

The study also emphasizes the role of **Business Analytics** in healthcare decision-making. Business Analytics involves the collection, processing, analysis, and visualization of operational

data to generate meaningful insights. Analytical tools such as SQL, Excel, Python, and Power BI are used to analyze hospital data, create dashboards, identify patterns, and support data-driven managerial decisions. These tools help hospital management improve operational planning, resource allocation, and service quality.

Thus, the theoretical framework integrates concepts of process management, resource optimization, workforce management, queue analysis, technology adoption, and business analytics to analyze the internal operations of Imperial Hospital. The framework provides a foundation for understanding operational inefficiencies and developing data-driven recommendations to improve hospital efficiency, workflow management, and patient care services.

1.4 Significance of the Project

This project is significant because it focuses on analyzing and improving the internal operations of hospitals through the application of Business Analytics techniques. Efficient hospital operations are essential for providing quality healthcare services, reducing patient waiting time, improving staff productivity, and ensuring the proper utilization of medical resources. In today's healthcare environment, hospitals face increasing operational challenges due to rising patient load, limited resources, and complex workflows. Therefore, the study helps in understanding these challenges and finding practical solutions through data-driven analysis.

The project is important for Imperial Hospital as it provides insights into different operational areas such as patient flow management, admission and discharge processes, staff workload distribution, resource utilization, and technology usage. By identifying bottlenecks and inefficiencies, the study can help hospital management improve operational performance and service quality.

The study also highlights the importance of Business Analytics in healthcare management. Through the use of tools such as SQL, Excel, Python, and Power BI, the project demonstrates how operational data can be transformed into meaningful insights, reports, dashboards, and visualizations for better decision-making. This helps hospital administrators make informed decisions related to staffing, scheduling, resource allocation, and workflow improvement.

The significance of the project can be understood through the following points:

- Helps identify operational inefficiencies and bottlenecks in hospital processes.
- Improves understanding of patient flow and waiting time management.
- Supports better staff workload distribution and shift planning.
- Assists in efficient utilization of hospital resources such as beds, ICU facilities, medicines, and equipment.
- Enhances coordination between different hospital departments.
- Demonstrates the practical application of Business Analytics in the healthcare sector.
- Supports data-driven decision-making and operational planning.
- Improves service quality and patient satisfaction.
- Helps reduce delays in admission, billing, discharge, and other operational activities.

- Provides recommendations for improving hospital workflow and efficiency.
- Contributes to academic research in healthcare analytics and hospital management systems.
- Enhances knowledge and practical understanding of analytical tools such as SQL, Excel, Python, and Power BI.

Overall, this project contributes to improving healthcare operations by combining hospital management practices with Business Analytics techniques. The study provides valuable insights that can help hospitals achieve better operational efficiency, resource management, and patient care outcomes.

1.5 Definition of terms

1. Business Analytics

Business Analytics refers to the process of collecting, analyzing, and interpreting data to support decision-making and improve organizational performance.

2. Hospital Management System (HMS)

A Hospital Management System is a system used to manage hospital activities such as patient registration, billing, appointment scheduling, and medical records.

3. Patient Flow

Patient flow refers to the movement of patients through different hospital processes such as registration, consultation, admission, treatment, and discharge.

4. Operational Efficiency

Operational efficiency refers to the ability of a hospital to perform its operations effectively with minimum delays and optimal use of resources.

5. Resource Utilization

Resource utilization means the effective use of hospital resources such as beds, ICU facilities, medicines, equipment, and staff.

6. Workforce Management

Workforce management involves planning and managing hospital staff, workload distribution, and shift scheduling to improve operational performance.

7. Queue Theory

Queue Theory is a concept used to analyze waiting lines and service delays, especially related to patient waiting time in hospitals.

8. Data Visualization

Data visualization refers to presenting data in graphical forms such as charts, graphs, dashboards, and reports for easier understanding and analysis.

CHAPTER NO 2
REVIEW OF LITERATURE

The review of literature provides an understanding of previous studies related to hospital management systems, healthcare operations, and the application of Business Analytics in the healthcare sector. It helps in understanding how analytical techniques and operational management practices can improve hospital efficiency, patient care, resource utilization, and decision-making processes. Various researchers have studied operational challenges in hospitals, including patient flow management, staff workload, waiting time, resource allocation, technology adoption, and healthcare service quality.

Review of Existing Literature Available on the Topic

1. Sood et al. (2017) analyzed the flow of operations in healthcare systems and highlighted the complexity of hospital processes involving multiple stakeholders. The study emphasized that inefficiencies in coordination and resource allocation can lead to increased operational delays and reduced service quality.
2. Mishra and Sathyaseelan (2019) identified key challenges in hospital management, such as staff shortages, inadequate infrastructure, and lack of awareness regarding efficient healthcare practices. They suggested that improving workforce planning and strengthening operational systems can enhance overall hospital performance.
3. Chaudhari et al. (2025) examined staff behavior and workload distribution in healthcare organizations. Their findings revealed that unequal workload, poor shift planning, and lack of proper scheduling systems significantly affect staff efficiency and patient care quality.
4. Gonce et al. (2017) proposed strategies to improve healthcare supply chains, including better demand forecasting, efficient inventory management, and use of digital tools. Their research supports the need for systematic planning and technological integration in hospital operations.
5. Prasad et al. (2024) focused on healthcare affordability and accessibility, highlighting the importance of efficient hospital management systems in reducing operational costs and improving patient outcomes.
6. Sharma and Gupta (2020) studied the role of Business Analytics in healthcare decision-making. The research concluded that data analytics tools help hospitals improve operational efficiency, monitor patient data, and support strategic planning.
7. Patil et al. (2021) analyzed patient flow management in hospitals and identified long waiting time, overcrowding, and poor departmental coordination as major operational challenges affecting healthcare services.
8. Kumar and Verma (2018) examined hospital resource utilization and found that improper bed management, equipment shortages, and inefficient inventory control negatively impact hospital performance and patient care quality.
9. Reddy et al. (2022) highlighted the importance of digital healthcare systems and hospital management software in improving data accuracy, communication, workflow efficiency, and operational control within hospitals.
10. Joshi and Kulkarni (2023) focused on healthcare service quality and patient satisfaction. The study emphasized that efficient hospital operations, reduced waiting time, and proper staff management are essential for improving patient experience and healthcare outcomes.

The reviewed studies indicate that efficient hospital management depends on proper coordination, workforce management, resource optimization, technology adoption, and data-driven decision-making. The literature also highlights the growing importance of Business Analytics in improving operational efficiency and healthcare service quality. Therefore, this study on Imperial Hospital aims to analyze internal hospital operations using Business Analytics techniques and provide practical recommendations for improving hospital efficiency and patient care services.

Conclusion of Literature Review

The review of literature shows that efficient hospital management is essential for improving operational performance, patient satisfaction, and healthcare service quality. Previous studies have highlighted major challenges such as patient delays, staff shortages, workload imbalance, inefficient resource utilization, poor coordination, and limited use of technology in hospital operations. Researchers have also emphasized the importance of Business Analytics, workforce management, patient flow analysis, and digital healthcare systems in improving operational efficiency and decision-making. The literature indicates that analytical tools and data-driven approaches help hospitals identify operational gaps, optimize resources, and enhance workflow management. However, limited studies focus specifically on analyzing internal hospital operations through a case-study-based Business Analytics approach. Therefore, this study on Imperial Hospital aims to analyze operational challenges and provide practical recommendations for improving hospital efficiency and service quality.

CHAPTER NO 3
COMPANY PROFILE

3.1 Introduction of Company

Imperial Hospital is a healthcare institution that provides medical and healthcare services to patients in Chikhali and nearby areas. The hospital offers facilities such as OPD, IPD, ICU, emergency care, diagnostic services, billing, and patient management services. With the support of doctors, nurses, and administrative staff, the hospital focuses on delivering quality healthcare services and efficient patient care. The hospital manages various operational activities including patient registration, consultation, admission, discharge, billing, and resource management. This study focuses on analyzing the internal operations of the hospital using Business Analytics techniques to improve operational efficiency, resource utilization, workflow management, and overall service quality.

3.2. Establishment and Background of the Organization

Imperial Hospital, located at Imperial Hospital, is a multi-speciality healthcare institution established to provide quality, affordable, and accessible medical services to patients in Chikhali and nearby areas of Pune. The hospital offers a wide range of healthcare services including dialysis, heart care, neurology, nephrology, gastroenterology, orthopaedics, trauma care, ICU, pathology, radiology, gynecology, pediatrics, and emergency services. The organization is supported by experienced doctors, nurses, technicians, and administrative staff who work together to ensure effective patient care and smooth hospital operations. With modern medical infrastructure, advanced diagnostic facilities, and specialized departments, the hospital focuses on improving healthcare quality, patient safety, and operational efficiency. The hospital also emphasizes the use of technology and organized management systems to handle patient registration, consultation, billing, admission, discharge, and resource management activities efficiently.

Vision

To provide quality, affordable, and patient-centered healthcare services with excellence and compassion.

Mission

To deliver advanced and efficient healthcare services through experienced professionals, modern technology, and quality patient care.

Core Values

Patient care, integrity, excellence, teamwork, innovation, commitment, and responsibility.

3.3. Nature of Business

Imperial Hospital operates in the healthcare and medical services sector. The hospital provides multispeciality healthcare services, diagnostic facilities, emergency care, and patient management services.

The hospital mainly focuses on:

- Providing outpatient and inpatient healthcare services
- Emergency and critical care treatment

- Diagnostic and laboratory services
- ICU and surgical care facilities
- Specialized medical treatments and consultations
- Patient admission, discharge, and billing services
- Healthcare support and resource management

The hospital also focuses on improving patient care quality, operational efficiency, and healthcare service delivery through modern medical infrastructure and healthcare management systems.

3.4. Services and Facilities Provided

Imperial Hospital provides various healthcare services including OPD, IPD, ICU, emergency care, diagnostic services, pathology, radiology, dialysis, cardiology, orthopaedics, gynecology, pediatrics, pharmacy, ambulance services, and patient management facilities with modern medical infrastructure and experienced healthcare professionals.

CHAPTER NO 4
RESEARCH METHODOLOGY

4.1 Introduction to Research & Research Design

4.1 Introduction to Research & Research Design

Research is a systematic process of collecting, analyzing, and interpreting information to understand a particular problem or topic. In this project, research is conducted to study and analyze the internal hospital operations of Imperial Hospital using Business Analytics techniques. The study focuses on understanding operational processes such as patient flow, staff workload, resource utilization, waiting time, and technology usage within the hospital.

The main objective of this research is to identify operational inefficiencies, understand the challenges faced by hospital staff, and analyze how hospital resources and processes are managed. The study also focuses on examining factors affecting operational efficiency and service quality in the hospital. The research is based on the opinions, experiences, and responses collected from doctors, nurses, and administrative staff working in the hospital.

This research is conducted with reference to the internal operational activities of Imperial Hospital, Chikhali, including departments such as OPD, IPD, ICU, Emergency, Billing, and Administration.

Research Design

Research design refers to the overall plan and structure used for conducting the research. It explains the methods used for collecting, measuring, and analyzing data.

The following research design is used for this study:

1) Type of Research

- Descriptive and Analytical Research

This study is descriptive in nature because it describes hospital operational processes, staff workload, patient flow, and resource management practices. The study is also analytical because it examines operational data to identify inefficiencies and improve decision-making.

2) Nature of Data

- Primary Data
- Secondary Data

The study is based on both primary and secondary data collected from various sources.

3) Data Collection Methods

- Survey Method
- Questionnaire Method

A structured questionnaire is used to collect information from doctors, nurses, and administrative staff regarding hospital operations, workload, patient flow, and resource management.

4) Sampling Method

- Convenience Sampling Method

Respondents are selected based on their availability and willingness to participate in the survey.

5) Sample Size

- The study includes responses collected from hospital staff members including doctors, nurses, and administrative employees. The study includes 100 respondents

Conclusion

The research design helps in the systematic collection and analysis of data related to internal hospital operations. It ensures that the study is organized, reliable, and useful for identifying operational challenges and providing data-driven recommendations for improving hospital efficiency and healthcare service quality.

4.2 Type of Research Undertaken

The research undertaken for this project is mainly descriptive and analytical research.

Descriptive Research

Descriptive research is used to describe and understand the existing operational processes, staff experiences, and management practices within the hospital. It helps in understanding:

- Patient flow and waiting time
- Staff workload and shift management
- Admission and discharge processes
- Resource utilization such as beds, ICU facilities, medicines, and equipment
- Technology usage and operational systems
- Challenges faced during daily hospital operations
- Overall operational efficiency and service quality

Analytical Research

Analytical research is used to analyze operational data, identify inefficiencies, and examine factors affecting hospital performance. It helps in identifying operational bottlenecks,

coordination gaps, workload imbalance, and areas requiring improvement through Business Analytics techniques.

Nature of Research-Primary Research

The study is mainly based on primary data collected directly from hospital staff through questionnaires, surveys, and observations. The respondents include doctors, nurses, and administrative staff working at Imperial Hospital who are involved in daily hospital operations and patient management activities.

Conclusion

Thus, the research undertaken in this project is descriptive and analytical in nature and is mainly based on primary data. The study helps in understanding the actual operational processes, staff experiences, and challenges faced during hospital management activities. It also supports the use of Business Analytics for improving operational efficiency, workflow management, and healthcare service quality.

4.3 Scope of the Project

The scope of the project defines the area and limitations of the study. This project focuses on analyzing the internal hospital operations of Imperial Hospital using Business Analytics techniques. The study mainly examines operational processes, staff management, patient flow, resource utilization, and technology usage within the hospital.

Scope of the Study

The study includes:

- Studying patient flow and waiting time management
- Understanding admission and discharge processes
- Analyzing staff workload and shift management
- Examining operational efficiency within different hospital departments
- Studying the utilization of hospital resources such as beds, ICU facilities, medicines, and equipment
- Understanding the use of technology and digital systems in hospital operations
- Identifying operational challenges and bottlenecks
- Analyzing coordination between departments and workflow management
- Generating insights using Business Analytics tools such as SQL, Excel, Python, and Microsoft Power BI
- Providing recommendations for improving hospital operations and service quality

Time Scope

- The study is conducted for a limited research period.

Data Scope

- The study is based on primary data collected through questionnaires, surveys, and observations from doctors, nurses, and administrative staff.
- Secondary data is collected from hospital records, reports, websites, and related research sources.

Limitations of Scope

- Limited sample size
- Study restricted to one hospital only
- Time constraints during data collection and analysis
- Some operational information may be confidential
- Results may not represent all hospitals or healthcare organizations

Conclusion

The scope of the project is limited but focused on understanding internal hospital operations and operational efficiency. The study helps in analyzing workflow management, resource utilization, staff workload, and patient management processes using Business Analytics techniques to improve healthcare service quality and hospital performance.

4.4 Different Sources & Methods of Data Collection

Data collection is an important part of research as it helps in gathering information required for analysis and interpretation. The study uses both primary and secondary sources of data to analyze the internal operations of Imperial Hospital.

1) Primary Data

Primary data refers to data collected directly from respondents for the first time.

Methods Used

a) Questionnaire Method

A structured questionnaire is used to collect responses from doctors, nurses, and administrative staff regarding hospital operations, workload, patient flow, and operational challenges.

b) Survey Method

A survey is conducted among hospital staff members to understand operational processes, workflow efficiency, staff workload, and the use of technology in hospital management.

Details of Primary Data

- Data collected from doctors, nurses, and administrative staff
- Information related to patient flow, waiting time, workload, resource management, and operational efficiency
- Data regarding technology usage and operational challenges within the hospital

2) Secondary Data

Secondary data refers to data already collected and available from different sources.

Sources of Secondary Data

- Hospital records and operational reports
- Research papers and journals
- Books and articles related to hospital management and Business Analytics
- Academic references and online study materials

Secondary data provides background knowledge, theoretical support, and understanding related to hospital operations and healthcare management systems.

Conclusion

Both primary and secondary data are important for the study. Primary data provides practical information about hospital operations, while secondary data supports research and theoretical understanding.

4.5 Sampling Design

Sampling design refers to the process of selecting respondents from the total population for data collection.

1) Target Population

The target population includes:

- Doctors
- Nurses
- Administrative staff

2) Sampling Unit

Each doctor, nurse, and administrative staff member working at Imperial Hospital is considered as a sampling unit.

3) Sampling Method

- Convenience Sampling Method

Respondents are selected based on their availability and willingness to participate in the study.

4) Sample Size

- The sample size consists of 100 respondents

5) Sampling Area

- Imperial Hospital, Chikhali, Pune

6) Data Collection Tool

- Structured Questionnaire

7) Characteristics of Sample

The sample includes:

- Male and female respondents
- Doctors, nurses, and administrative staff
- Employees from different departments such as OPD, IPD, ICU, Emergency, and Billing
- Staff members with different experience levels and work shifts

Conclusion

The sampling design helps in selecting suitable respondents for the study and ensures relevant data collection for analyzing hospital operations, staff workload, patient flow, and operational efficiency.

4.6 Selection of Proper Data Analysis Methods, Techniques, or Tools Used

In this project, statistical, analytical, and graphical methods are used for analyzing the data collected from hospital staff and operational records.

1) Percentage Method

- Used to convert responses into percentages
- Helps in easy understanding and interpretation of operational data

Examples:

- Percentage of staff facing workload issues
- Percentage of delays in patient admission or discharge
- Percentage of respondents satisfied with hospital operations

2) Graphical Method

Graphs and charts are used for visual presentation of data.

Types of Graphs Used

- Bar Charts
- Pie Charts
- Line Graphs
- Dashboard Visualizations

These graphical tools make the analysis simple, clear, and easy to understand.

3) Questionnaire Analysis

- Each response collected through questionnaires is analyzed separately
- Interpretation is done based on responses provided by doctors, nurses, and administrative staff

4) Business Analytics Tools Used

- SQL for data extraction and querying
- Excel for data cleaning and tabular analysis
- Python for data processing and analysis
- Microsoft Power BI for dashboards and data visualization

Conclusion

The use of percentage analysis, graphical presentation, and Business Analytics tools helps in effective interpretation of hospital operational data. These methods support better understanding of patient flow, staff workload, resource utilization, and operational efficiency within the hospital.

4.7 Necessary Assumptions

Assumptions are the conditions considered true for conducting the research smoothly.

The following assumptions are made in this study:

1. Respondents have provided honest and accurate responses.
2. Respondents have proper knowledge of hospital operations and management processes.
3. Doctors, nurses, and administrative staff have understood the questionnaire properly before answering.
4. The selected sample represents the operational conditions of the hospital.
5. Hospital operations and workflow remain stable during the study period.
6. The collected data is free from major bias and errors.
7. The information collected from respondents reflects actual operational practices within the hospital.

8. The hospital follows standard operational procedures for patient management and healthcare services.
9. The research is conducted under normal working conditions within the hospital environment.

Conclusion

These assumptions help in conducting the research effectively and support the process of data analysis and interpretation regarding internal hospital operations, operational efficiency, and healthcare management practices.

CHAPTER – 05
DATA ANALYSIS & INTERPRETATION

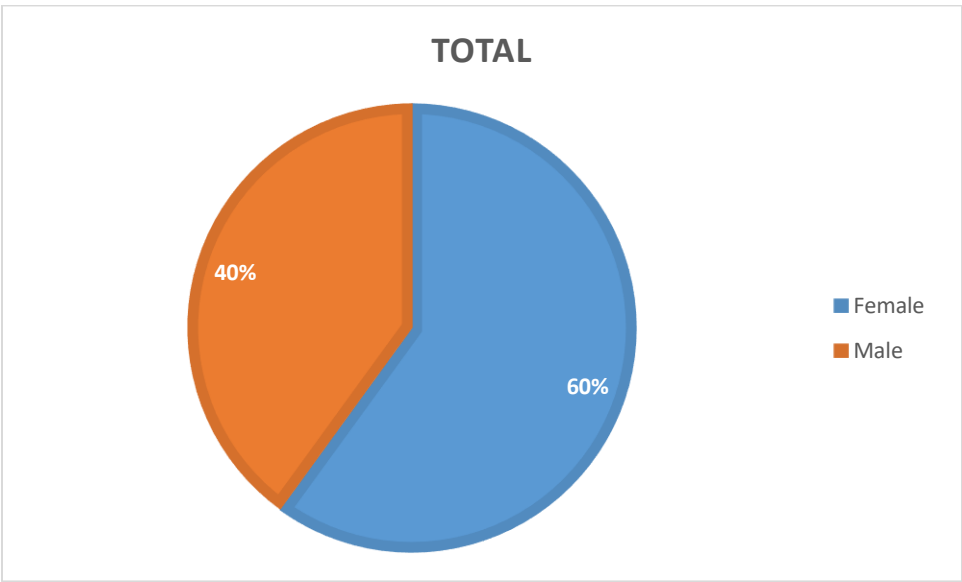
Section A: Demographic Information: -

Question 1: What is your gender?

Table 1.1: Gender-wise Distribution of Respondents

Gender	No of Respondent	Percentage
Male	40	40
Female	60	60
Total	100	100%

Figure 1.1 Gender-wise Distribution of Respondents



Summary

The above table and chart show the gender-wise distribution of respondents included in the study. A total of 100 respondents participated in the survey.

The data reveals that all respondents are male, representing 40% of the total sample, while 60% female respondents participated in the study.

Interpretation

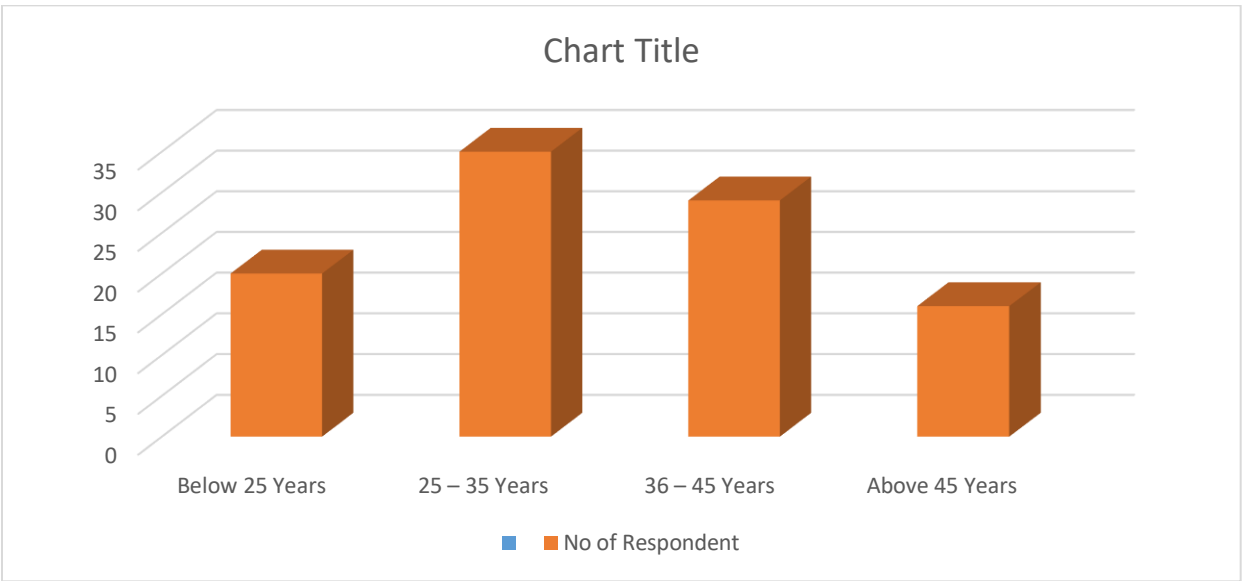
The above data shows that out of 100 respondents, 40% are male and 60% are female. This indicates that female respondents constitute the majority of the sample selected for the study. The higher participation of female respondents may be due to the greater involvement of female staff in hospital operations and healthcare services.

Question 2: Which age group do you belong to?

Table 1.2: Age-wise Distribution of Respondents

Age	No of Respondent	Percentage
Below 25 Years	20	20%
25 – 35 Years	35	35%
36 – 45 Years	29	29%
Above 45 Years	16	16%

Figure 1.2



Summary

The table shows the age-wise distribution of respondents in the study. The highest number of respondents belongs to the 25–35 years age group (35%), followed by 36–45 years (29%). The below 25 years group accounts for 20%, while the above 45 years group has the lowest

representation at 16%. The data indicates that most respondents are from the young to middle-aged working population.

Interpretation

The analysis indicates that the majority of respondents are in the productive working age group (25–45 years), which suggests that the responses are largely from experienced healthcare professionals or staff actively involved in hospital operations. The lower representation of the above 45 years group may indicate fewer senior employees or reduced participation from older staff members. This age distribution helps ensure that the study captures insights primarily from active workforce participants in Imperial Hospital, Chikhali.

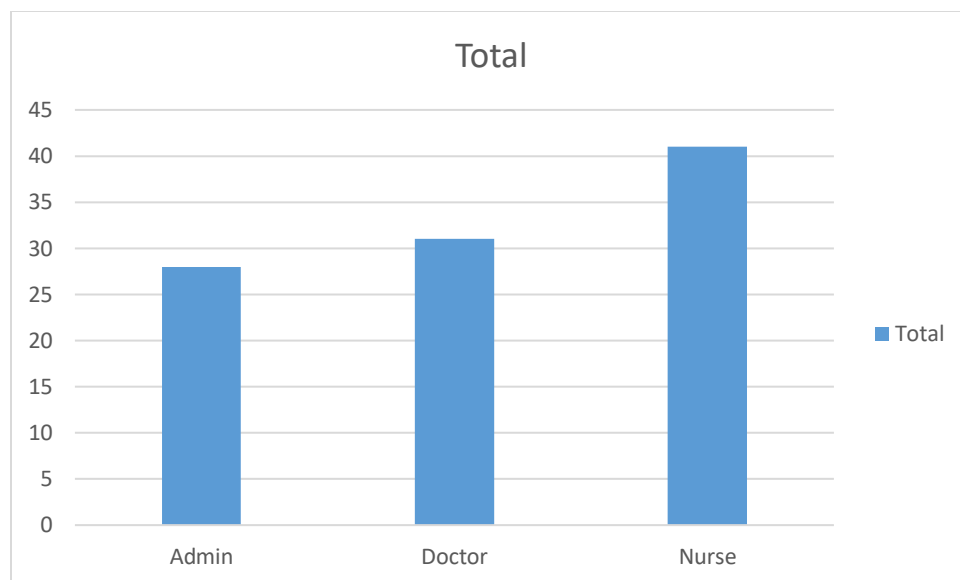
Section B – Internal Hospital Operations

Question 3: Please indicate your Role in the hospital (Admin/Doctor/Nurse).

Table 2.1: Role-wise Distribution of Respondents

Response	No of Respondent	Percentage
Admin	28	28%
Doctor	31	31%
Nurse	41	41%

Figure 2.1 Role-wise Distribution



Summary

The table represents the distribution of respondents based on their roles in the hospital. The highest proportion of respondents are Nurses (41%), followed by Doctors (31%), and Administrative staff (28%). This indicates that nursing staff form the largest group among the respondents.

Interpretation

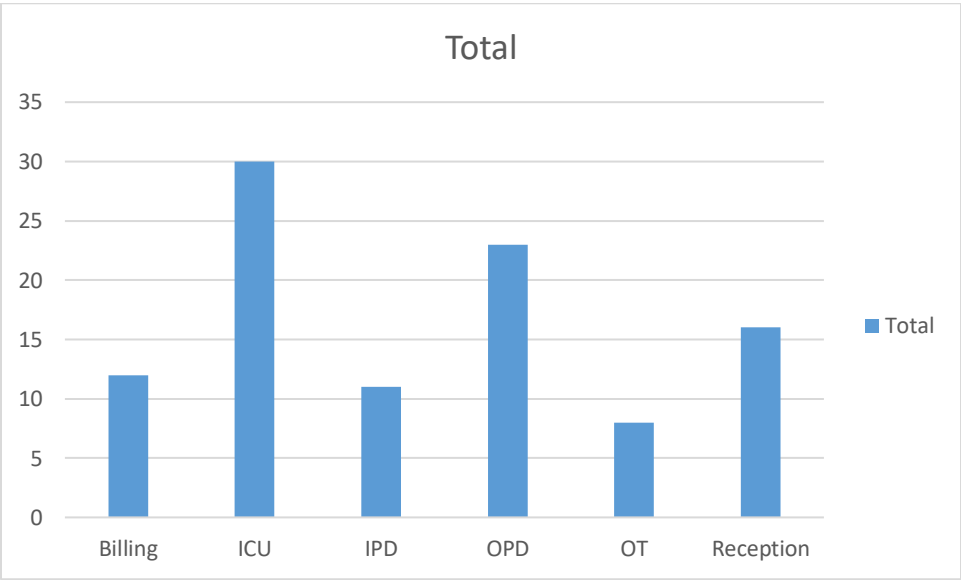
The data shows that nurses constitute the majority of the respondents, reflecting their significant involvement in day-to-day patient care and hospital operations at Imperial Hospital, Chikhali. Doctors also form a substantial portion, contributing professional and clinical perspectives to the study. Administrative staff have comparatively lower representation, but they provide important insights into operational and support functions. Overall, the response distribution ensures a balanced view of hospital operations from clinical as well as administrative departments.

Question 4: Please indicate your Department in the hospital

Table 2.2: Department-wise Distribution of Respondents

Row Labels	Count of Respondent-ID
Billing	12
ICU	30
IPD	11
OPD	23
OT	8
Reception	16
Grand Total	100

Figure 2.2



Summary

The table shows the distribution of respondents across different hospital departments. The highest number of respondents is from the ICU (30), followed by OPD (23) and Reception (16). Billing (12) and IPD (11) have moderate representation, while the OT (8) has the lowest number of respondents.

Interpretation

The data indicates that ICU and OPD staff form a major part of the respondents, highlighting their active involvement in patient care and hospital operations. Reception and billing departments also contribute significantly, reflecting their role in administrative and patient

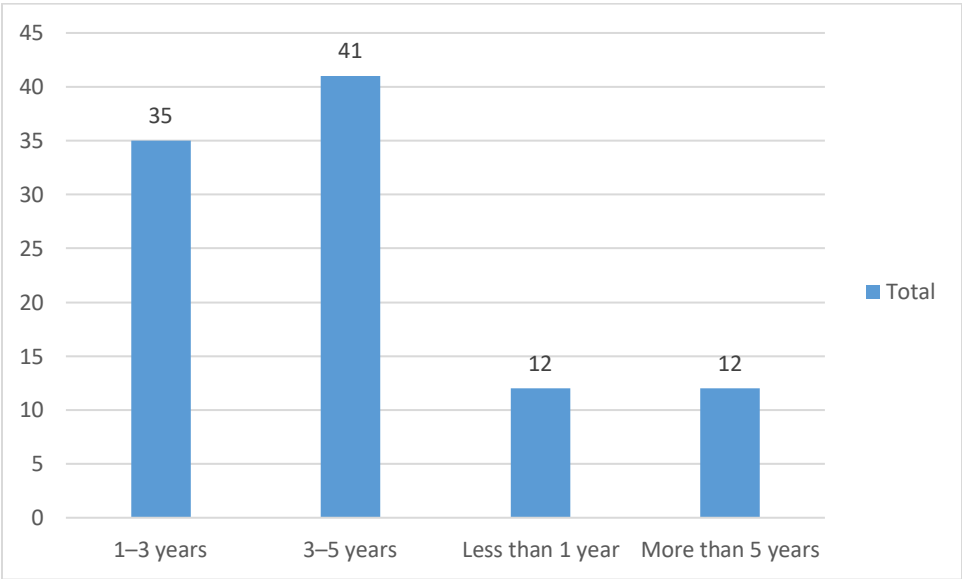
support services. The lower representation from OT and IPD suggests comparatively fewer respondents from specialized or inpatient services. Overall, the distribution provides a balanced view of both clinical and administrative departments at Imperial Hospital, Chikhali.

Question 5: Could you please mention your experience?

Table 2.3: Experience Distribution of Hospital Staff

Row Labels	Count of Respondent_ID
1–3 years	35
3–5 years	41
Less than 1 year	12
More than 5 years	12
Grand Total	100

Figure 2.3 Experience Distribution



Summary

The table shows the distribution of respondents based on their work experience. The highest number of respondents have 3–5 years of experience (41%), followed by 1–3 years (35%). Respondents with less than 1 year (12%) and more than 5 years (12%) are equally represented in smaller proportions. The total number of respondents is 100.

Interpretation

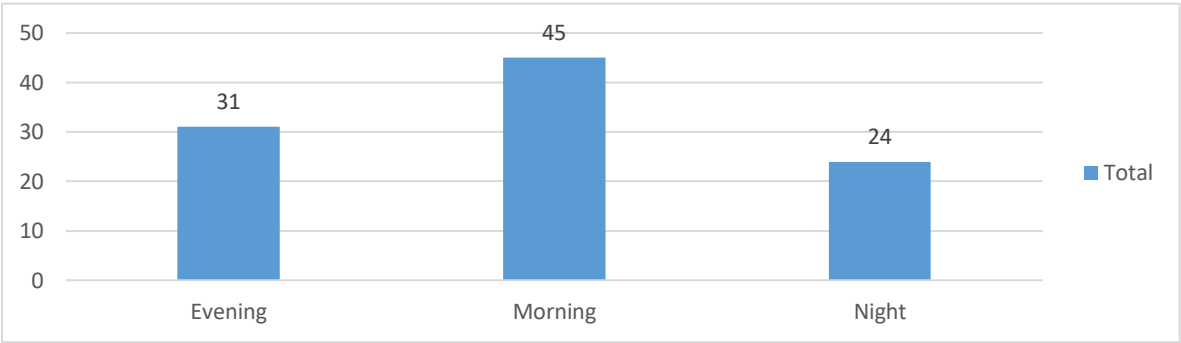
The data indicates that most respondents fall in the early to mid-career stage (1–5 years of experience), suggesting that the study primarily reflects the views of moderately experienced staff. The equal representation of very new and highly experienced employees provides additional balance, but their lower proportion may limit insights from extreme experience levels. Overall, the dataset captures perspectives mainly from actively engaged and regularly working hospital staff at Imperial Hospital, Chikhali.

Question 6: Could you please mention your Shift times?

Table 2.4: Shift-wise Distribution of Respondents

Row Labels	Count of Respondent_ID
Evening	31
Morning	45
Night	24
Grand Total	100

Figure 2.4



Summary of Data

The table presents the distribution of respondents across different work shifts. The Morning shift (45%) has the highest number of respondents, followed by the Evening shift (31%). The Night shift (24%) has the lowest representation. The total number of respondents is 100.

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Interpretation

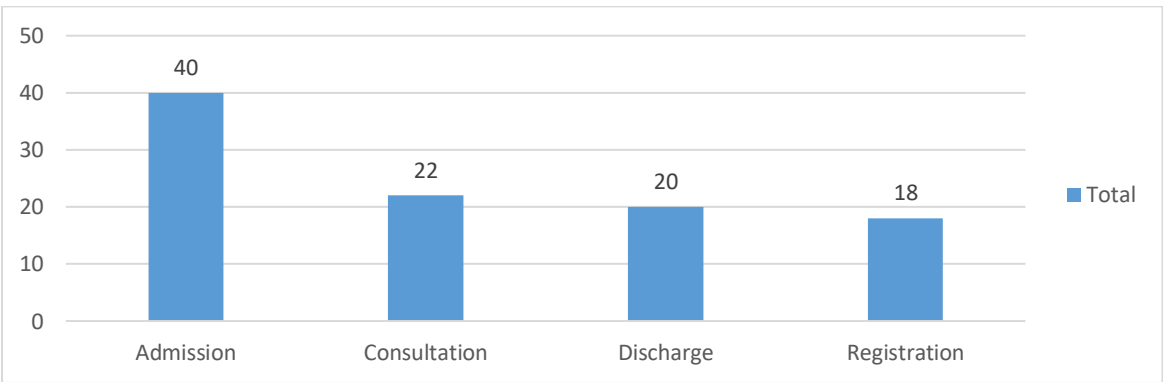
Most respondents are from the morning shift, indicating higher staffing and activity during daytime operations. Evening and night shifts have lower representation, showing comparatively reduced workforce presence during those hours at Imperial Hospital, Chikhali.

Question 7: At which stage do patients face maximum delay?

Table 2.5 Stage-wise Patient Delay Analysis in Hospital Operations

Row Labels	Count of Respondent_ID
Admission	40
Consultation	22
Discharge	20
Registration	18
Grand Total	100

Figure 2.5:-



Summary of Data

The table shows the stages where patients face maximum delays during hospital operations. The highest number of respondents identified **Admission (40%)** as the stage with maximum delay, followed by **Consultation (22%)**, **Discharge (20%)**, and **Registration (18%)**. The total number of respondents is 100.

Interpretation

The data indicates that the admission process is the most time-consuming stage in hospital operations at Imperial Hospital, Chikhali. This may be due to documentation, bed allocation, or administrative procedures. Consultation and discharge also contribute to patient waiting time, while registration shows comparatively fewer delays. Overall, the findings highlight the need to improve efficiency in the admission process to reduce patient waiting time and enhance service quality.

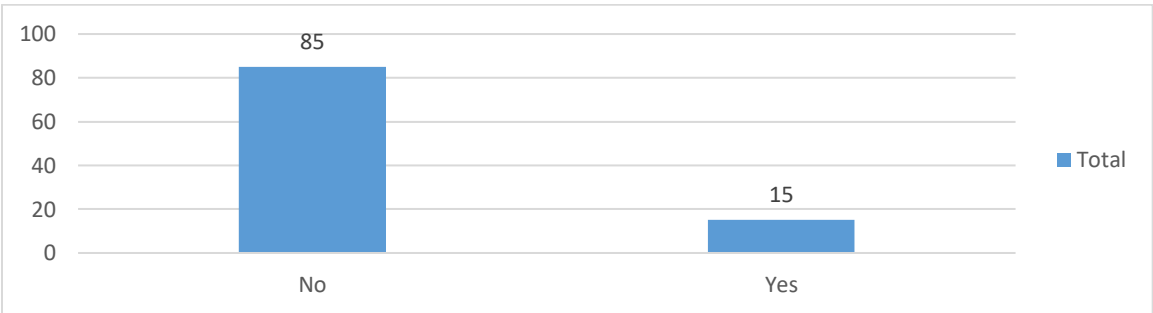
Section C – Staff & Workload Management

Question 8: Is staff sufficient during peak hours?

Table 3.1: Availability of Staff During Peak Hours

Row Labels	Count of Respondent_ID
No	85
Yes	15
Grand Total	100

Figure 3.1:



Summary of Data

The table presents respondents’ opinions regarding staff sufficiency during peak hours. A majority of respondents (85%) stated that staff is not sufficient during peak hours, while only 15% believe that staffing is sufficient. The total number of respondents is 100.

Interpretation

The data clearly indicates that most respondents feel there is a shortage of staff during peak working hours at Imperial Hospital, Chikhali. This shortage may lead to increased workload, longer patient waiting times, and operational inefficiencies. The findings suggest a need for better workforce planning and staff allocation to improve hospital service efficiency during busy

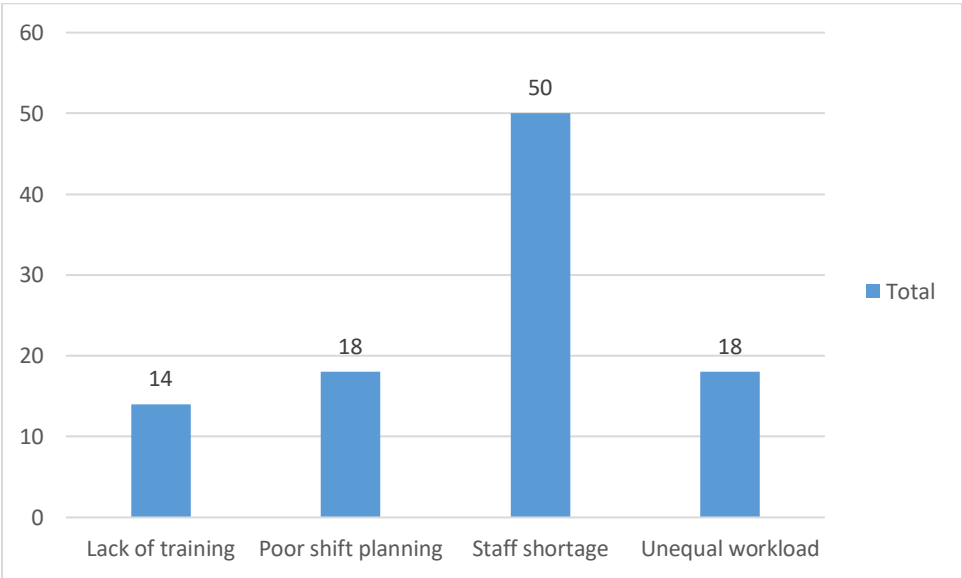
periods.

Question 9: What is the major staff-related issue?.

Table 3.2: Major Staff-related Issues in Hospital Operations

Row Labels	Count of Respondent_ID
Lack of training	14
Poor shift planning	18
Staff shortage	50
Unequal workload	18
Grand Total	100

Figure 3.2:



Summary of Data

The table shows the major staff-related issues faced in hospital operations. The majority of respondents identified Staff shortage (50%) as the main issue. Poor shift planning (18%) and Unequal workload (18%) were reported equally, while Lack of training (14%) was the least reported issue. The total number of respondents is 100.

Interpretation

Staff shortage is the major issue affecting hospital operations, leading to increased workload and reduced efficiency. Poor shift planning and unequal workload also impact staff performance, highlighting the need for better workforce management at Imperial Hospital, Chikhali.

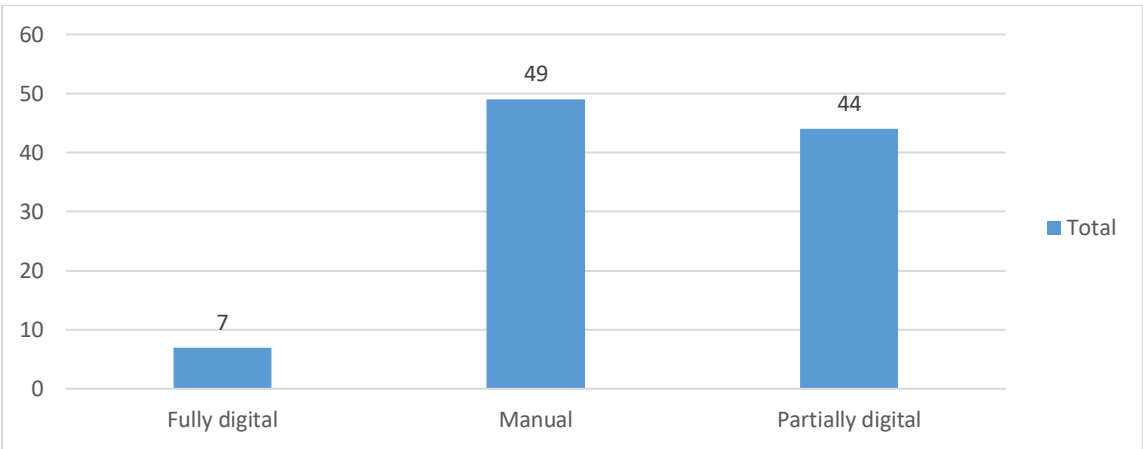
Section D: Technology & Systems

Question 10 Current system used in hospital:.

Table 3.3: Current System Used in Hospital Operations

Row Labels	Count of Respondent_ID
Fully digital	7
Manual	49
Partially digital	44
Grand Total	100

Figure 3.3 :



Summary of Data

The table shows the type of system currently used in hospital operations. The majority of respondents reported the use of a **Manual system (49%)**, followed by a **Partially digital system (44%)**. Only **7%** of respondents stated that the hospital uses a **Fully digital system**. The total number of respondents is 100

Interpretation

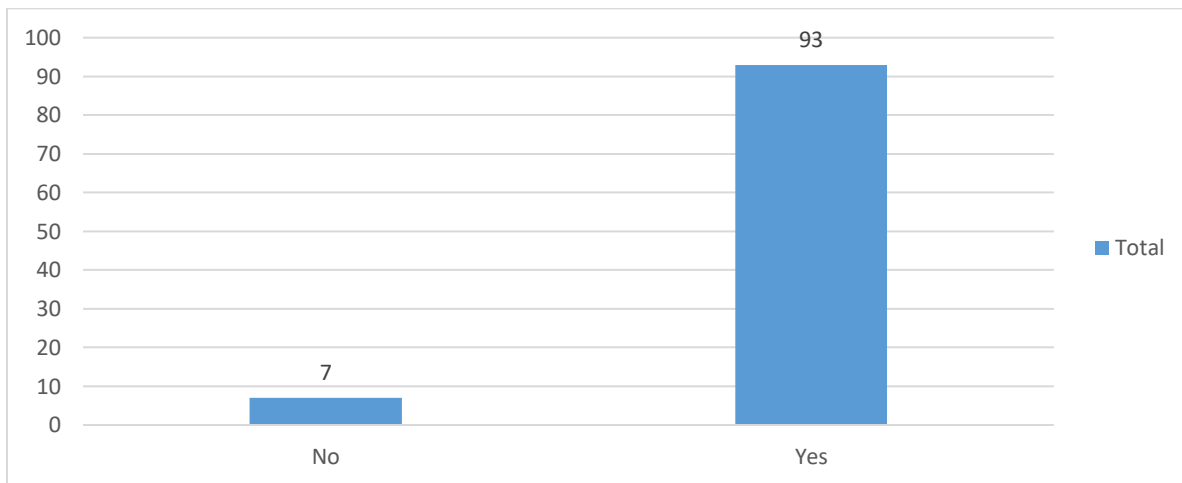
The data indicates that hospital operations at Imperial Hospital, Chikhali are still largely dependent on manual processes, with limited adoption of fully digital systems. A significant proportion of partially digital systems suggests that digital transformation has started but is not yet fully implemented. The low percentage of fully digital operations highlights the need for greater technological adoption to improve efficiency, accuracy, and operational management.

Question 11:- . Do system/software issues affect your work?

Table 3.4: Impact of System/Software Issues on Work Efficiency

Row Labels	Count of Respondent_ID
No	7
Yes	93
Grand Total	100

Figure 3.4:



Summary of Data

The table shows respondents' views on whether system or software issues affect their work. A large majority of respondents (**93%**) stated that system/software issues affect their work, while only **7%** reported no impact.

Interpretation

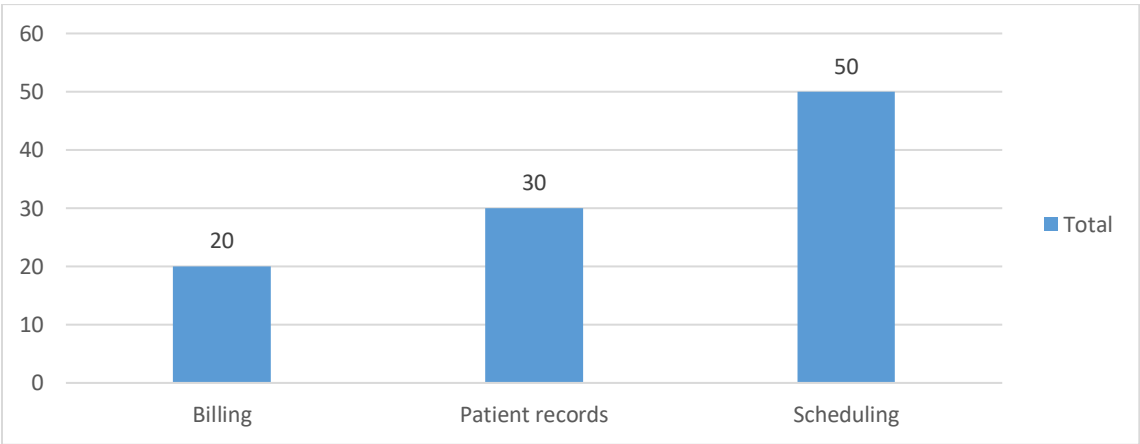
The data indicates that system and software-related issues significantly affect hospital operations at Imperial Hospital, Chikhali. Frequent technical problems may reduce work efficiency, delay processes, and increase staff workload. The findings highlight the importance of improving system reliability and technical support to ensure smoother hospital operations.

Question 12: Which area needs improvement through technology?

Table 3.5:- Areas Needing Improvement Through Technology

Row Labels	Count of Respondent_ID
Billing	20
Patient records	30
Scheduling	50
Grand Total	100

Figure 3.5:



Summary of Data

The table shows the hospital areas that require improvement through technology. The majority of respondents identified Scheduling (50%) as the main area needing improvement, followed by Patient records (30%) and Billing (20%)

Interpretation

Scheduling is the main area requiring technological improvement, as inefficient scheduling can cause delays and workload imbalance. Patient records and billing also need better digital systems to improve operational efficiency at Imperial Hospital, Chikhali.

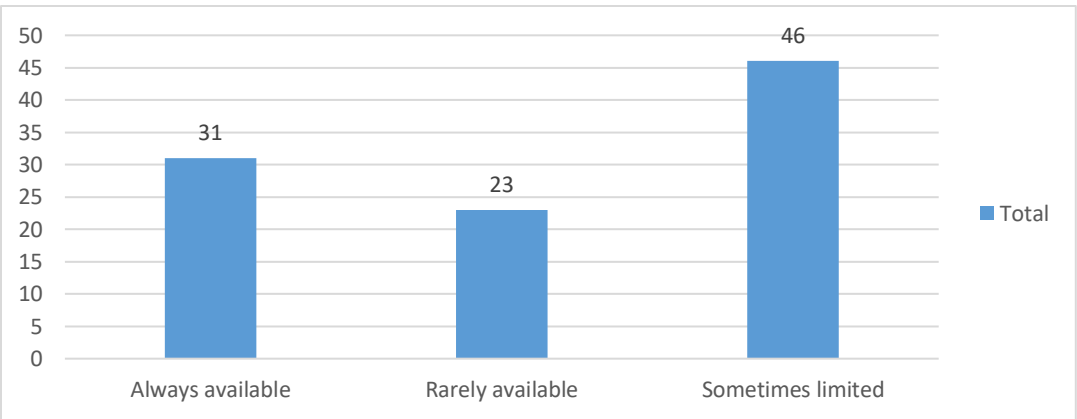
Section E: Resource & Infrastructure

Question 13: ICU bed availability:

Table 3.5:- Availability of ICU Beds in Hospital Operations

Row Labels	Count of Respondent_ID
Always available	31
Rarely available	23
Sometimes limited	46
Grand Total	100

Figure 3.5:



Summary of Data

The table shows the availability of ICU beds in the hospital. The highest number of respondents reported that ICU beds are Sometimes limited (46%). Around 31% stated that beds are Always available, while 23% reported that beds are Rarely available.

Interpretation

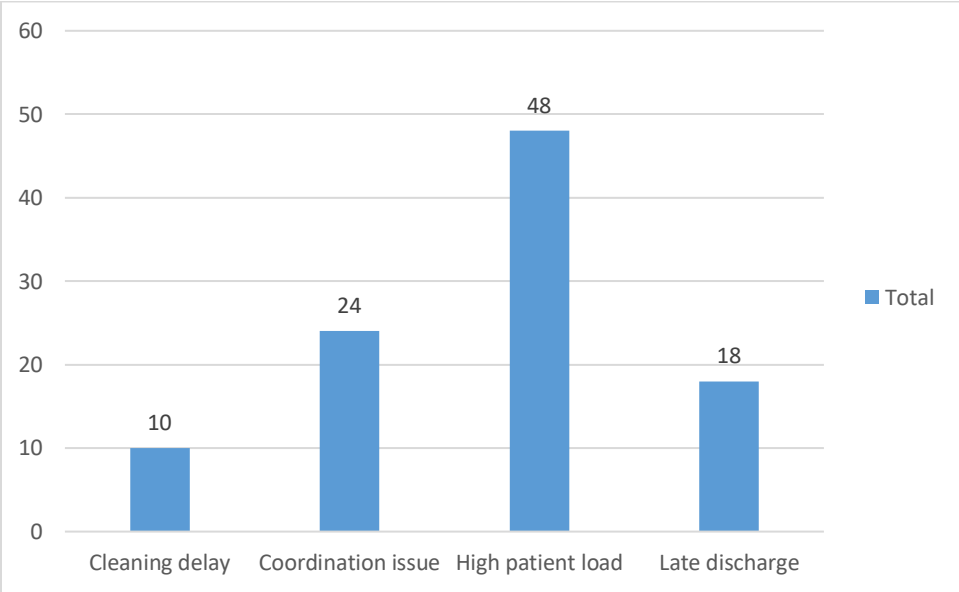
The data indicates that ICU bed availability is inconsistent at Imperial Hospital, Chikhali, with many respondents experiencing limited availability at times. This may affect emergency care and patient management during high-demand periods. The findings suggest the need for better resource planning and capacity management to improve ICU services.

Question 13: Main reason for delay in bed allocation:

Table 3.5:- Major Reasons for Delay in Bed Allocation

Row Labels	Count of Respondent_ID
Cleaning delay	10
Coordination issue	24
High patient load	48
Late discharge	18

Figure 3.5:



Summary

The table shows the main reasons for delays in bed allocation at the hospital. The majority of respondents identified High patient load (48%) as the primary reason. This is followed by Coordination issues (24%), Late discharge (18%), and Cleaning delay (10%).

Interpretation

The data indicates that high patient load is the major factor causing delays in bed allocation at Imperial Hospital, Chikhali. Coordination issues and delayed discharge processes also contribute significantly to the problem, while cleaning delays have a comparatively lower impact. The findings highlight the need for better patient flow management and coordination to improve bed availability and operational efficiency.

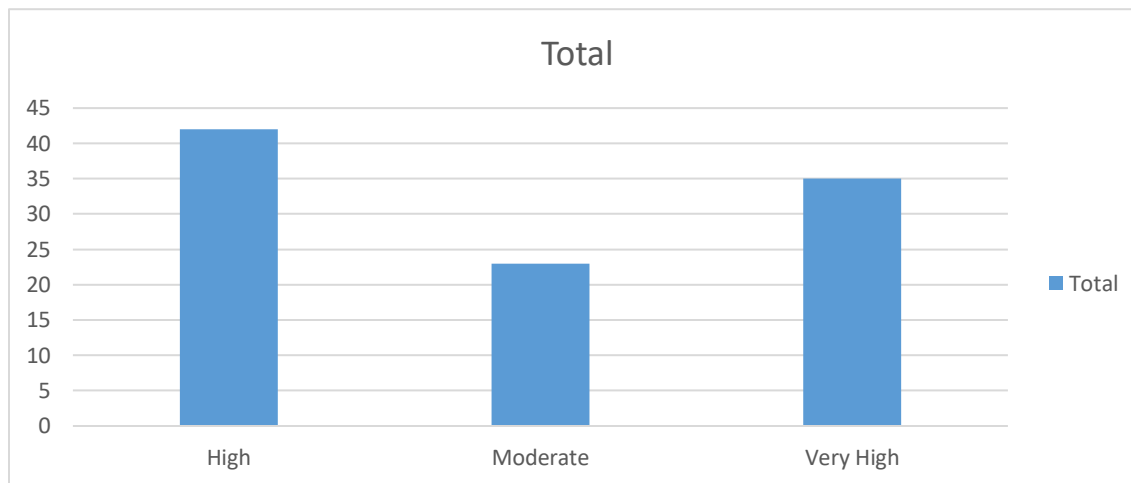
Section F: Patient-Related Operations & Experience

Question 14: How would you describe daily patient load?

Table : Daily Patient Load in Hospital Operations

Row Labels	Count of Respondent_ID
High	42
Moderate	23
Very High	35
Grand Total	100

Figure 3.5:



Summary

The table shows respondents' views on the daily patient load in the hospital. The highest number of respondents described the patient load as High (42%), followed by Very High (35%) and Moderate (23%). The total number of respondents is 100.

Interpretation

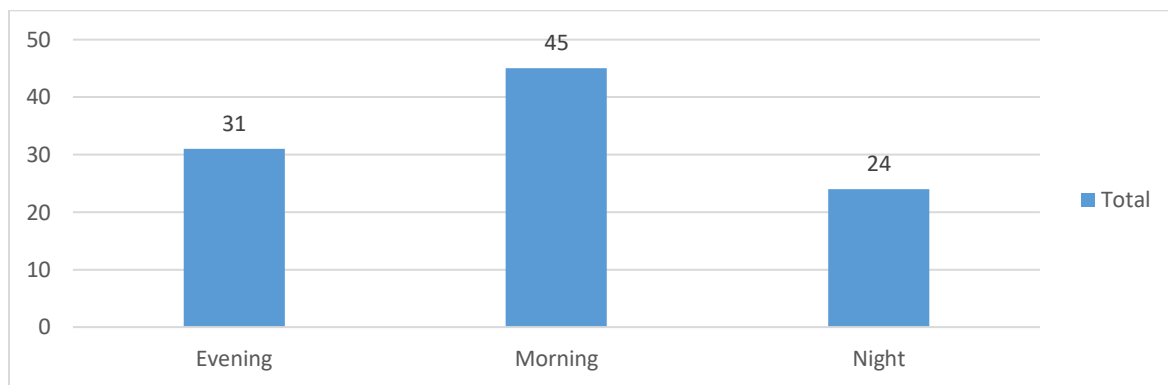
The data indicates that Imperial Hospital, Chikhali experiences a consistently high patient load. A significant proportion of respondents also reported a very high patient load, which may increase pressure on hospital resources, staff, and operational efficiency. The findings suggest the need for effective workload management and resource planning to handle patient demand efficiently.

Question 15: During which time is patient load highest?

Table : Time-wise Analysis of Patient Load

Row Labels	Count of Respondent_ID
Evening	31
Morning	45
Night	24
Grand Total	100

Figure 3.5



Summary

The table shows the time during which patient load is highest in the hospital. The majority of respondents reported the Morning shift (45%) as having the highest patient load, followed by the Evening shift (31%) and the Night shift (24%).

Interpretation

The data indicates that patient load is highest during the morning hours at Imperial Hospital, Chikhali. This may be due to outpatient consultations, admissions, and routine medical services being concentrated during the daytime. Evening and night shifts experience comparatively lower patient flow, though they still contribute significantly to hospital operations.

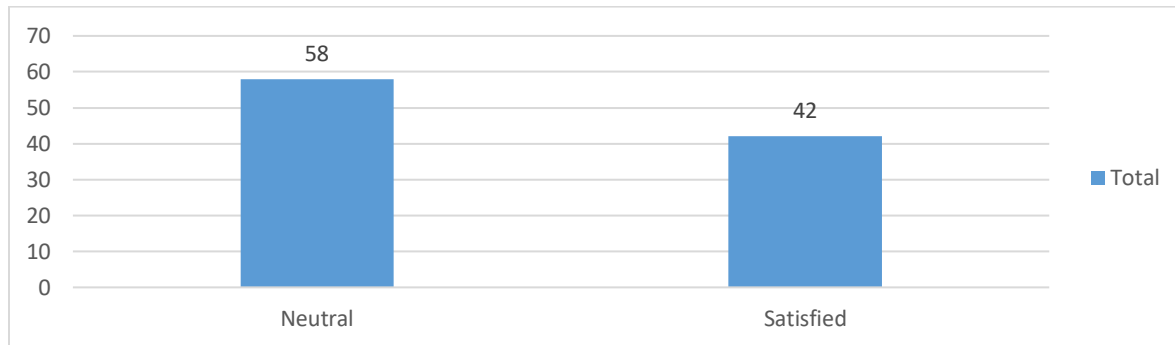
Section G: Overall Management

Question 16: How satisfied are you with internal hospital management?

Table: Satisfaction Level with Internal Hospital Management

Row Labels	Count of Respondent_ID
Neutral	58
Satisfied	42
Grand Total	100

Figure 3.5



Summary

The table shows respondents' satisfaction levels regarding internal hospital management. A majority of respondents are Neutral (58%), while 42% reported being Satisfied with the hospital management system.

Interpretation

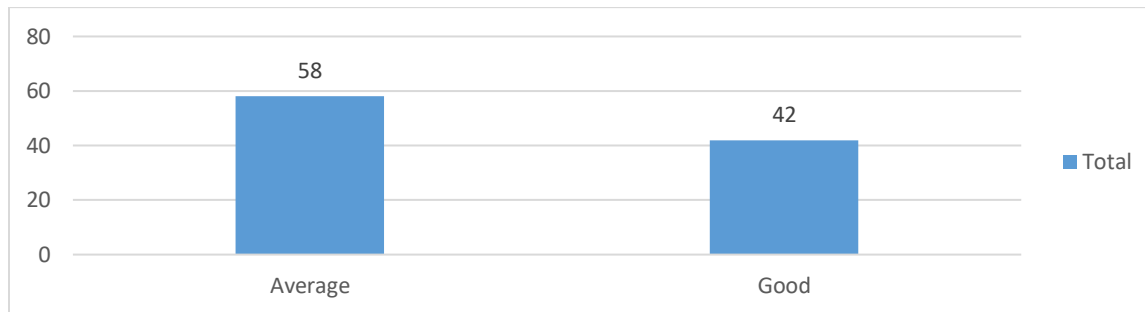
The data indicates that most respondents have a neutral opinion about internal hospital management at Imperial Hospital, Chikhali. While a considerable number of respondents are satisfied, the high neutral response suggests there is scope for improvement in management practices, communication, and operational efficiency.

Question 17: Overall, how would you rate operational efficiency of the hospital?

TableOverall Rating of Hospital Operational Efficiency

Row Labels	Count of Respondent_ID
Average	58
Good	42
Grand Total	100

Figure 3.5



Summary

The table presents respondents' ratings of the hospital's operational efficiency. The majority of respondents rated the operational efficiency as Average (58%), while 42% rated it as Good. The total number of respondents is 100.

Interpretation

The data indicates that most respondents consider the operational efficiency of Imperial Hospital, Chikhali to be average. Although a significant proportion rated it as good, the higher average rating suggests that there is room for improvement in workflow management, resource utilization, and service delivery to enhance overall hospital performance.

CHAPTER NO 6
FINDING & SUGGESTIONS

6.1 Findings

Based on the data collected and analysed, the following key findings are observed:

Findings of the Study

1. Age-wise Distribution of Respondents

Most respondents belong to the 25–35 years age group (35%), followed by the 36–45 years category (29%).

The study mainly represents the views of young and middle-aged hospital employees.

2. Role-wise Distribution of Respondents

Nurses form the highest proportion of respondents (41%), followed by doctors (31%) and administrative staff (28%).

Clinical staff participation is higher, reflecting their active involvement in hospital operations.

3. Department-wise Distribution of Respondents

ICU (30%) and OPD (23%) departments have the highest respondent participation in the study. The findings include responses from both clinical and administrative departments.

4. Experience-wise Distribution of Respondents

Most respondents have 3–5 years of work experience (41%), followed by 1–3 years (35%).

The study mainly reflects the opinions of moderately experienced employees.

5. Shift-wise Distribution of Respondents

The morning shift has the highest number of respondents (45%) compared to evening (31%) and night shifts (24%).

Hospital activities and staff involvement are mainly concentrated during morning hours.

6. Stage with Maximum Delay

Admission is identified as the stage where patients face maximum delay (40%), followed by consultation (22%).

Improving admission procedures can help reduce waiting time and improve service quality.

7. Staff Sufficiency During Peak Hours

Most respondents (85%) believe staff is insufficient during peak working hours.

Staff shortage may affect operational efficiency and patient service quality.

8. Major Staff-related Issues

Staff shortage is the major staff-related issue reported by respondents (50%).
Poor shift planning (18%) and unequal workload (18%) also affect employee performance.

9. Current System Used in Hospital

The hospital mainly depends on manual (49%) and partially digital systems (44%).
The low use of fully digital systems (7%) shows limited technological adoption.

10. Impact of System/Software Issues

Most respondents (93%) stated that system or software issues affect their work.
Technical problems may reduce efficiency and delay hospital operations.

11. Areas Needing Improvement Through Technology

Scheduling is the main area requiring technological improvement (50%), followed by patient records (30%).
Better digital systems can improve coordination and operational efficiency.

12. ICU Bed Availability

Many respondents reported that ICU beds are sometimes limited (46%).
Inconsistent bed availability may affect emergency patient care and treatment.

13. Main Reason for Delay in Bed Allocation

High patient load is the major reason for delays in bed allocation (48%).
Better coordination and patient flow management can reduce such delays.

14. Daily Patient Load

Most respondents described the daily patient load as high (42%) or very high (35%).
Heavy patient flow may increase pressure on hospital staff and resources.

15. Time of Highest Patient Load

The morning shift experiences the highest patient load (45%) in the hospital.
Additional staff and resource planning may be required during peak hours.

16. Satisfaction with Internal Hospital Management

Most respondents have a neutral opinion regarding internal hospital management (58%).
The findings indicate scope for improvement in management practices and communication.

17. Overall Operational Efficiency Rating

Most respondents rated the hospital's operational efficiency as average (58%), while 42% rated it as good.

The hospital requires improvements in workflow and resource management for better performance.

6.2 Suggestions

Based on the findings, the following suggestions are recommended:

- ☐ The hospital should increase staff availability during peak hours to reduce workload pressure and improve patient care.
- ☐ Digital systems should be implemented for scheduling, billing, and patient record management to improve operational efficiency.
- ☐ Admission and bed allocation processes should be streamlined to reduce patient waiting time.
- ☐ Regular maintenance and upgradation of software systems should be carried out to avoid technical issues.
- ☐ Better shift planning and workload distribution should be adopted to improve employee performance.
- ☐ ICU bed management should be improved to ensure availability during emergencies.
- ☐ Training programs should be conducted regularly to improve staff skills and system handling.
- ☐ Hospital management should focus on improving coordination among departments for smoother operations.
- ☐ Resource planning should be strengthened to manage high patient load effectively.

- ☐ Feedback from employees and patients should be considered to improve overall hospital operational efficiency.

CHAPTER 7
REFLECTION

7.1 Reflection

This study helped in understanding the internal operational challenges faced by Imperial Hospital, Chikhali. It provided practical knowledge about hospital workflow, staff management, patient handling, and the importance of technology in healthcare operations.

The research highlighted that staff shortages, high patient load, delays in admission and bed allocation, and software-related issues affect operational efficiency. Through business analytics, the study identified key problem areas and suggested improvements that can help enhance hospital performance and patient satisfaction.

The study also improved analytical, communication, and research skills while providing real-world exposure to healthcare management practices. Overall, the project was a valuable learning experience that demonstrated the importance of data-driven decision-making in hospital operations.

During the project, I faced certain challenges such as difficulty in collecting responses, lack of awareness among some respondents, and time management during data collection and analysis. These challenges helped me improve my problem-solving abilities and adaptability.

Another key learning was understanding customer behaviour. It was observed that trust, convenience, and simplicity are the major factors influencing the adoption of digital payment systems.

CHAPTER 8
CONCLUSION

Conclusion

The study titled “Analyzing Internal Hospital Operations Using Business Analytics: A Case Study of Imperial Hospital, Chikhali” was conducted to understand and evaluate the internal operational efficiency of the hospital. The research focused on key operational areas such as patient flow, staff management, hospital systems, bed allocation, shift management, and the impact of technology on healthcare services.

The findings of the study revealed that the hospital faces several operational challenges, including staff shortages during peak hours, high patient load, delays in admission and bed allocation, and software-related issues. Most respondents indicated that scheduling and patient record management require technological improvement. The study also found that the hospital still depends significantly on manual and partially digital systems, which may affect efficiency, coordination, and service quality.

The analysis further showed that patient load is highest during morning shifts, increasing pressure on staff and hospital resources. ICU bed availability was reported as inconsistent, and delays in bed allocation were mainly caused by high patient inflow and coordination issues. These factors directly influence patient satisfaction and the smooth functioning of hospital operations.

Through the application of business analytics, the study successfully identified critical operational gaps and areas requiring improvement. The research emphasized the importance of better workforce planning, efficient resource utilization, improved coordination among departments, and adoption of advanced digital systems to streamline hospital processes.

The study concludes that the implementation of technology-driven solutions, effective staff management, and data-based decision-making can significantly improve operational efficiency and patient care services at Imperial Hospital, Chikhali. Overall, the project provided valuable insights into healthcare operations and demonstrated how business analytics can support better management practices and organizational performance in hospitals.

CHAPTER 9
REFERENCE

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- Imperial Hospital, Chikhali – Internal records and operational data collected during the study.
- Articles and journals related to hospital operations, healthcare analytics, and hospital management referred to during the research study.
- Microsoft Excel and Power BI tools were used for data analysis and graphical representation.

- Various online resources and research papers related to business analytics in healthcare management were referred to for academic understanding and project development.

Appendix

The survey questionnaire used for primary data collection is presented below for reference

